

Arannav Saxena

AI Engineer | Generative AI Engineer

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Profile Summary

AI Engineer with experience in building Agentic workflows, Retrieval-Augmented Generation (RAG) systems, machine learning pipelines, scalable backend services and a real-time Voice AI application. Skilled in Python and libraries like PyTorch, LangChain, LangGraph, vector databases, and LLM observability using LangSmith. Interested in designing reliable, real-time AI infrastructure and production-ready applications.

Experience

Generative AI Engineer Intern – eSaarthi - Stealth AI startup

Nov 2025 – Apr 2026

- Designed multi-agent state machines using LangGraph with conditional routing and tool-calling loops, achieving above 90% accuracy in automated context retrieval.
- Engineered hybrid RAG pipelines on Qdrant with parent-document chunking, reducing hallucinations by around 30% on internal evaluation datasets.
- Built asynchronous FastAPI services with bidirectional WebSocket endpoints, streaming token responses across 150 concurrent sessions at under 2s average latency.
- Integrated STT/TTS audio AI modules with custom preprocessing pipelines and established LangSmith telemetry, reducing voice-query inference latency and cutting chain failure debugging time by 40%.
- Contributed to a personalized AI educational assistant for students, collaborating via Git workflows with feature branching and shared deployments.

Education

B.Tech in Computer Science Engineering (AIML), Ajay Kumar Garg Engineering College, Ghaziabad

2027

CGPA: 7.80/10.00

Technical Skills

- Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, SQL
- Data/ML:** PyTorch, NumPy, Pandas, Scikit-learn, XGBoost, Deep Learning, Machine Learning
- Languages:** Python, C++ , C
- GenAI / LLM Stack:** LangChain, LangGraph, LlamaIndex, RAG, Agentic Workflows, Tool Calling, Prompt Engineering
- LLMOps:** LangSmith, Evaluation, Tracing, Monitoring
- Backend:** FastAPI, Flask, REST APIs, WebSockets
- Databases:** Qdrant, PostgreSQL, MongoDB, SQLite
- Infra & Tools:** Docker, Git, Linux, Colab
- Audio AI:** Librosa, Mel Spectrograms
- Voice AI:** Telephony (Twilio), Speech processing, Streaming Audio, STT and TTS models, VAD (Voice Activity Detection), WebSockets, LiveKit

Key Projects

Call Handling System

- Built a modular real-time voice agent from the ground up using an asynchronous state machine, custom Silero VAD, streaming Cartesia STT/TTS, and Groq LLM, achieving sub-2s response latency across 500+ interactions.
- Designed an AudioDevice abstraction with swappable backends (local mic/speaker and Twilio telephony), enabling barge-in via VAD speech detection and real-time μ -law/PCM audio conversion.
- Implemented a FastAPI/Twilio server with dynamic TwiML generation and a finite-state machine (IDLE → LISTENING → PROCESSING → SPEAKING → INTERRUPTED) for robust call flow control.

AI Education Assistant

- Built an adaptive AI-powered learning assistant tailored to individual student learning styles.
- Integrated LangChain, LangGraph, and LlamaIndex to orchestrate multi-step reasoning workflows.
- Implemented RAG for accurate, context-aware responses using curated knowledge bases.
- Leveraged Supermemory to maintain personalized learning context across sessions.

Raga Detector

- Developed a CNN-based classifier for Indian classical music raga recognition.
- Built a dataset from audio clips of multiple artists and generated mel spectrograms using Librosa.
- Trained a Convolutional Neural Network (CNN) on mel spectrograms to classify two Indian classical ragas.

Achievements

- Participated in 5+ Kaggle competitions with practical exposure to regression modeling and feature engineering.
- Regular competitive programmer solving algorithmic challenges in C++ on LeetCode and CodeChef.

Courses

Machine Learning – Andrew Ng (Coursera)

Anthropic API course

Pandas – Kaggle